

Dilation-like identities for recurrences over semirings: three extensions

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In our paper submission, we considered recurrences over semirings and how to vectorize them. In particular, we developed two transformations that we collectively call “dilation”: one works with rings, and the other works with idempotent semirings. In both cases, we use algebraic identities to transform an input recurrence into one whose recurrence kernel has a larger stride. A larger stride allows more values of a recurrence to be computed in parallel, enabling vectorization.

However, a broader class of dilation-like identities can be derived from other properties of the semiring. In this post, I will show three extensions of the dilation-like identities from the paper. Each one assumes an additional property of the semiring:

1. The semiring is a zero-divisor-free ring with a primitive m -th root of unity. This generalizes the paper’s ring dilation.
2. The semiring is a ring of prime characteristic p . This specializes the ring setting.
3. The semiring is c -saturating. An example of a c -saturating semiring is saturating arithmetic. This also generalizes the paper’s dilation for idempotent semirings.

What is a dilation-like identity?

Let us first define what a dilation-like identity is. Consider a recurrence of the form

$$F = \text{Rec}(K, G), \tag{1}$$

for a causal, time-invariant kernel K and a signal G . This recurrence computes the fixed-point solution to $F = K \otimes F \oplus G$. Throughout this post, we only consider commutative semirings, so that kernels commute under convolution.

Suppose all nonzero taps of K occur at offsets divisible by s . We say that K has stride s . A dilation-like identity has the form

$$\boxed{\text{Rec}(K, G) = \text{Rec}(L, P \otimes G)} \quad (2)$$

If the stride of L is ms , we say that Equation (2) is a factor- m **dilation**, and we call P the **prefilter**.

It is helpful to consider dilation-like identities over rings. For rings, the above identity is equivalent to the following:

$$\begin{aligned} I \ominus L &= P \otimes (I \ominus K) \\ L &= I \ominus P \otimes (I \ominus K) \end{aligned} \quad (3)$$

To see this, note that convolution with $I \ominus K$ is the inverse operation of the recurrence $\text{Rec}(K, \cdot)$. All kernels in this discussion are time-invariant, so convolutions and recurrences commute.

Therefore, finding a dilation-like identity over a ring largely reduces to finding an appropriate prefilter P for which L is causal and has a larger stride.

In the paper, we used two dilation-like identities.

- When the semiring is a ring, write a kernel with stride s as $A \oplus B$, where A contains the taps at offsets divisible by $2s$ and B contains the taps at offsets congruent to s modulo $2s$. We have

$$\text{Rec}(A \oplus B, G) = \text{Rec}(A \oplus A \ominus A^2 \oplus B^2, (I \ominus A \oplus B) \otimes G). \quad (4)$$

Every term in the new recurrence kernel, A , A^2 , and B^2 , has taps only at multiples of $2s$, so the new kernel has stride $2s$.

- When the recurrence is over an idempotent semiring, also known as a dioid, we have

$$\text{Rec}(A \oplus B, G) = \text{Rec}(A \oplus B^2, (I \oplus B) \otimes G). \quad (5)$$

With a slight generalization,

$$\text{Rec}\left(\bigoplus_{i=1}^n A_i, G\right) = \text{Rec}\left(\bigoplus_{i=1}^n A_i^m, \bigotimes_{i=1}^n \left(\bigoplus_{j=0}^{m-1} A_i^j\right) \otimes G\right). \quad (6)$$

Let each A_i be a singleton kernel containing one tap of $\bigoplus_{i=1}^n A_i$. Replacing A_i by A_i^m multiplies its offset by m , while the prefilter accounts for the intermediate powers $I, A_i, \dots, A_i^{m-1}$.

The following sections show a broader class of identities that generalize these two identities.

Extension 1: Zero-divisor-free rings with primitive m -th roots of unity

Let us consider the case in which the algebraic structure is a ring with a primitive m -th root of unity. We say a value ω is an m -th root of unity if $\omega^m = 1_\tau$. A ring has $\ominus_\tau 1_\tau$ as a second root of unity, since $(\ominus_\tau 1_\tau)^2 = 1_\tau$. An m -th root of unity ω is *primitive* if $\omega^r \neq 1_\tau$ for every integer r with $1 \leq r < m$. An example is the ring of complex numbers, where, for every m ,

$$\omega = e^{2\pi i/m}$$

is a primitive m -th root of unity: raising it to the m -th power gives 1. Note that the second root of unity $\ominus_\tau 1_\tau$ is not necessarily primitive, because some rings satisfy $\ominus_\tau 1_\tau = 1_\tau$.

Let K be a causal, time-invariant kernel with stride s . Define a transformation ρ on such kernels by

$$\rho(K)[ns] = \omega^n K[ns]. \quad (7)$$

In other words, $\rho(K)$ multiplies every tap of K at offset ns by ω^n . We have $\rho(K_1 \oplus K_2) = \rho(K_1) \oplus \rho(K_2)$ and $\rho(K_1 \otimes K_2) = \rho(K_1) \otimes \rho(K_2)$. Denote j repeated applications of ρ by ρ^j ; then $\rho^j(K)[ns] = \omega^{jn} K[ns]$. Because $\omega^m = 1_\tau$, the m -th application of ρ returns the original kernel: $\rho^m = \text{id}$.

Suppose the ring satisfies the implication $a \otimes_\tau b = 0_\tau \Rightarrow a = 0_\tau$ or $b = 0_\tau$; such a ring is called **zero-divisor-free**. Then K is a fixed point of ρ (that is, $\rho(K) = K$) if and only if every nonzero tap of K occurs at an offset divisible by ms . The backward direction follows from $\omega^m = 1_\tau$. For the forward direction, suppose $\rho(K) = K$. Then for every n ,

$$\begin{aligned} K[ns] &= \omega^n K[ns] \\ (1_\tau \ominus_\tau \omega^n) \otimes_\tau K[ns] &= 0_\tau. \end{aligned}$$

Together, the zero-divisor-free assumption and the primitivity of ω imply that either n is divisible by m or $K[ns] = 0_\tau$.

We can now define two finite kernels:

$$P_m(K) = \bigotimes_{j=1}^{m-1} (I \ominus \rho^j(K)) \quad (8)$$

and

$$L_m(K) = I \ominus \bigotimes_{j=0}^{m-1} (I \ominus \rho^j(K)). \quad (9)$$

Because the kernels are time-invariant, their convolutions commute, so Equations (8) and (9) give

$$\begin{aligned} I \ominus L_m(K) &= \bigotimes_{j=0}^{m-1} (I \ominus \rho^j(K)) \\ &= P_m(K) \otimes (I \ominus K). \end{aligned}$$

This is exactly the identity in Equation (3), so

$$\boxed{\text{Rec}(K, G) = \text{Rec}(L_m(K), P_m(K) \otimes G).} \quad (10)$$

The new recurrence kernel $L_m(K)$ is causal: expanding the product in Equation (9) gives I plus terms supported on positive offsets, so $L_m(K)$ has no tap at offset 0. Moreover, $L_m(K)$ is a fixed point of ρ , because

$$\begin{aligned} \rho(L_m(K)) &= \rho \left(I \ominus \bigotimes_{j=0}^{m-1} (I \ominus \rho^j(K)) \right) \\ &= I \ominus \bigotimes_{j=0}^{m-1} \rho(I \ominus \rho^j(K)) \\ &= I \ominus \bigotimes_{j=1}^m (I \ominus \rho^j(K)) \\ &= L_m(K). \end{aligned}$$

Therefore, if the ring is zero-divisor-free, $L_m(K)$ has stride ms .

(Side note suggested by ChatGPT and Claude: in algebraic language, ρ is the automorphism of the offset-graded kernel algebra induced by the $\mathbb{Z}/m$ grading, and $\bigotimes_{j=0}^{m-1} \rho^j(\cdot)$ is the norm of the cyclic group generated by ρ . The norm is ρ -invariant, which is exactly the argument above.)

Recovering the paper's ring dilation

Take $m = 2$, so $\omega = \ominus_\tau 1_\tau$ is a second root of unity. Write $K = A \oplus B$, where A contains the even-offset taps and B the odd-offset taps. Then

$$\rho(K) = A \ominus B.$$

Equations (8) and (9) become

$$P_2(K) = I \ominus A \oplus B$$

and

$$\begin{aligned} L_2(K) &= I \ominus (I \ominus A \ominus B) \otimes (I \ominus A \oplus B) \\ &= A \oplus A \ominus A^2 \oplus B^2. \end{aligned}$$

This recovers the paper's dilation for recurrences over rings. Note that deriving the paper's binary ring dilation requires neither the zero-divisor-free assumption nor the primitivity of $\ominus_\tau 1_\tau$. Those assumptions were only used to establish the implication that $\rho(K) = K$ forces K to have stride ms .

Example: tripling the stride of a real recurrence

Take $m = 3$ and $\omega = e^{2\pi i/3}$, and let $K = [0, a, b]$ be a kernel with stride $s = 1$ and real taps. Then $\rho(K) = [0, \omega a, \omega^2 b]$ and $\rho^2(K) = [0, \omega^2 a, \omega b]$, and Equations (8) and (9) give

$$\begin{aligned} P_3(K) &= [1, a, a^2 + b, -ab, b^2], \\ L_3(K) &= [0, 0, 0, a^3 + 3ab, 0, 0, b^3]. \end{aligned}$$

The new recurrence kernel has taps only at offsets 3 and 6, as promised. Note also that both results are real, even though $\rho(K)$ has complex taps. This is not luck: for every m , if all taps of K are reals, then so do $P_m(K)$ and $L_m(K)$. To see this, write $\bar{X}$ for the kernel obtained by replacing every tap of X with its complex conjugate, so that X has real taps exactly when $\bar{X} = X$, and note that conjugation respects $\oplus$ and $\otimes$. Because $\bar{\omega} = \omega^{m-1}$, a kernel K with real taps satisfies $\overline{\rho^j(K)} = \rho^{m-j}(K)$, so conjugation only permutes the factors of the products in Equations (8) and (9). Hence $\overline{P_m(K)} = P_m(K)$ and $\overline{L_m(K)} = L_m(K)$: their taps are real for every m , and the root of unity is only used as a device for the derivation.

Extension 2: Rings of prime characteristic

Suppose that the sum of p copies of 1_τ is zero:

$$\underbrace{1_\tau \oplus_\tau \cdots \oplus_\tau 1_\tau}_{p \text{ copies}} = 0_\tau. \quad (11)$$

The least positive number of copies for which this happens is called the **characteristic**. For example, arithmetic modulo 3 has characteristic 3:

$$1 + 1 + 1 = 0 \pmod{3}.$$

If p copies of 1_τ add to zero, then for any coefficient a ,

$$\underbrace{a \oplus_\tau \cdots \oplus_\tau a}_{p \text{ copies}} = \left(\underbrace{1_\tau \oplus_\tau \cdots \oplus_\tau 1_\tau}_{p \text{ copies}} \right) \otimes_\tau a = 0_\tau.$$

Therefore the sum of $p - 1$ copies of a is an additive inverse of a : every semiring of finite characteristic is already a ring, so finite characteristic is a special case of ring arithmetic.

What changes when the characteristic is prime?

Assume now that the characteristic p is prime. For any time-invariant kernels X and Y and any exponent $q \geq 1$, the binomial theorem gives

$$(X \oplus Y)^q = \bigoplus_{r=0}^q \binom{q}{r} X^r \otimes Y^{q-r}.$$

Now take $q = p^h$ with $h \geq 1$. Every intermediate binomial coefficient $\binom{q}{r}$, for $0 < r < q$, is then divisible by p , so Equation (11) makes every mixed term vanish, leaving

$$(X \oplus Y)^q = X^q \oplus Y^q. \tag{12}$$

This is the **Frobenius identity**.

In the paper, we have the following unrolling identity for any semiring:

$$\text{Rec}(K, G) = \text{Rec}\left(K^q, \bigoplus_{i=0}^{q-1} K^i \otimes G\right). \tag{13}$$

Normally, this identity is not useful for dilation because K^q expands to mixed products of taps of K and does not have a larger stride.

Prime characteristic removes exactly those mixed products. Write the time-invariant kernel K as a sum of singleton kernels, one for each tap of K :

$$K = \bigoplus_{i=1}^n K_i.$$

Taking $q = p^h$ and applying the Frobenius identity gives

$$K^q = \bigoplus_{i=1}^n K_i^q.$$

Thus, in a ring of prime characteristic p , for any $q = p^h$ with $h \geq 1$,

$$\boxed{\text{Rec}(K, G) = \text{Rec}\left(\bigoplus_{i=1}^n K_i^q, \bigoplus_{i=0}^{q-1} K^i \otimes G\right)} \quad (14)$$

If K originally has stride s , the new feedback kernel has stride qs .

Example: arithmetic modulo 3

Consider the ring of integers modulo 3. It has prime characteristic $p = 3$. Let the kernel be $K = [0, 2, 1]$. We have

$$\begin{aligned} K^3 &= [0, 0, 0, 2^3, 0, 0, 1] \\ &= [0, 0, 0, 2, 0, 0, 1], \\ I \oplus K \oplus K^2 &= [1] \oplus [0, 2, 1] \oplus [0, 0, 1, 1, 1] \\ &= [1, 2, 2, 1, 1]. \end{aligned}$$

The recurrence can therefore be rewritten as

$$\text{Rec}(K, G) = \text{Rec}([0, 0, 0, 2, 0, 0, 1], [1, 2, 2, 1, 1] \otimes G).$$

The stride of the feedback kernel has been tripled.

This result applies naturally to modular arithmetic and other rings of prime characteristic. Because finite characteristic already implies the ring properties, this transformation can be viewed as a refinement of the paper's ring setting.

Extension 3: c -saturating semirings

Fix an integer $c \geq 1$. For a nonnegative integer n , write

$$[n]_\tau = \underbrace{1_\tau \oplus_\tau \cdots \oplus_\tau 1_\tau}_{n \text{ copies}}, \quad [0]_\tau = 0_\tau.$$

We will say that a semiring is c -saturating if

$$[c]_\tau = [c + 1]_\tau. \quad (15)$$

Adding more copies of 1_τ to both sides shows that

$$[n]_\tau = [c]_\tau \quad \text{when } n \geq c. \quad (16)$$

Multiplying both sides by any value a gives

$$\underbrace{a \oplus_\tau \cdots \oplus_\tau a}_{n \text{ copies}} = \underbrace{a \oplus_\tau \cdots \oplus_\tau a}_{c \text{ copies}} \quad (n \geq c). \quad (17)$$

In other words, addition saturates at c copies of any value. A natural example of this is saturation arithmetic capped at c .

When $c = 1$, Equation (15) says $1_\tau = 1_\tau \oplus_\tau 1_\tau$, which is equivalent to $a = a \oplus_\tau a$ for every a . This recovers (additive) idempotence. Thus c -saturation is a generalization of idempotence.

As an aside, the database literature has studied p -stability as a sufficient condition for the convergence of Datalog programs over semirings. A semiring is said to be **p -stable** when

$$1_\tau \oplus_\tau a \oplus_\tau \cdots \oplus_\tau a^p = 1_\tau \oplus_\tau a \oplus_\tau \cdots \oplus_\tau a^p \oplus_\tau a^{p+1} \quad (18)$$

for every coefficient a . Setting $a = 1_\tau$ gives the condition for $(p+1)$ -saturation. Thus every p -stable semiring is $(p+1)$ -saturating. The converse, however, does not hold in general.

Deriving the prefilter

Over an idempotent semiring, a path generated twice counts the same as a path generated once, so the prefilter in Equation (5) only has to supply the powers of B that the dilated kernel skips. Over a c -saturating semiring, multiplicities below c are still visible, so the prefilter must repair those as well.

Let A and B be causal, time-invariant kernels, and fix an integer $m \geq 1$. We look for a dilation-like identity whose dilated form is $\text{Rec}(A \oplus B^m, P_{m,c}(A, B) \otimes G)$, where the prefilter $P_{m,c}(A, B)$ is defined in terms of A and B . Our goal is to find such a prefilter. Below, $\langle n \rangle$ denotes the kernel with a single tap of semiring value $[n]_\tau$ at offset 0, so that multiplying a kernel by $\langle n \rangle$ adds n copies of it.

$$\begin{aligned} \text{Rec}(A \oplus B, G) &= \text{Rec}(A \oplus B^m, P_{m,c}(A, B) \otimes G) \\ \text{Rec}(A \oplus B, G) &= (A \oplus B^m) \otimes \text{Rec}(A \oplus B^m, P_{m,c}(A, B) \otimes G) \oplus P_{m,c}(A, B) \otimes G \\ \text{Rec}(A \oplus B, G) &= (A \oplus B^m) \otimes \text{Rec}(A \oplus B, G) \oplus P_{m,c}(A, B) \otimes G \\ \bigoplus_{i,j \geq 0} \left\langle \begin{pmatrix} i+j \\ j \end{pmatrix} \right\rangle \otimes A^i \otimes B^j \otimes G &= \left(P_{m,c}(A, B) \oplus \bigoplus_{i,j \geq 0} (A \oplus B^m) \otimes \left\langle \begin{pmatrix} i+j \\ j \end{pmatrix} \right\rangle \otimes A^i \otimes B^j \right) \otimes G \end{aligned}$$

Since we want this to hold for every G , it suffices that the two kernels agree:

$$\begin{aligned}
\bigoplus_{i,j \geq 0} \left\langle \binom{i+j}{j} \right\rangle \otimes A^i \otimes B^j &= P_{m,c}(A, B) \oplus \bigoplus_{i,j \geq 0} (A \oplus B^m) \otimes \left\langle \binom{i+j}{j} \right\rangle \otimes A^i \otimes B^j \\
&= P_{m,c}(A, B) \\
&\quad \oplus \bigoplus_{i,j \geq 0} \left\langle \binom{i+j}{j} \right\rangle \otimes A^{i+1} \otimes B^j \\
&\quad \oplus \bigoplus_{i,j \geq 0} \left\langle \binom{i+j}{j} \right\rangle \otimes A^i \otimes B^{j+m} \\
&= P_{m,c}(A, B) \oplus \bigoplus_{i,j \geq 0} \left\langle \mathbf{1}_{i>0} \binom{i+j-1}{j} + \mathbf{1}_{j \geq m} \binom{i+j-m}{j-m} \right\rangle \otimes A^i \otimes B^j
\end{aligned}$$

Let $N_{i,j} = \binom{i+j}{j}$ and $R_{m,i,j} = \mathbf{1}_{i>0} \binom{i+j-1}{j} + \mathbf{1}_{j \geq m} \binom{i+j-m}{j-m}$, so that $N_{i,j}$ and $R_{m,i,j}$ are the multiplicities of $A^i \otimes B^j$ on the left- and right-hand sides of the equation above. Note that

$$\begin{aligned}
N_{i,j} = \binom{i+j}{j} &= \binom{i+j-1}{j} + \binom{i+j-1}{j-1} \\
&\geq \binom{i+j-1}{j} + \binom{i+j-m}{j-m} \geq R_{m,i,j}.
\end{aligned}$$

The following definition of $P_{m,c}(A, B)$ satisfies the above equation:

$$\begin{aligned}
P_{m,c}(A, B) &= \bigoplus_{i,j \geq 0} \langle D_{m,i,j} \rangle \otimes A^i \otimes B^j \\
\text{where } D_{m,i,j} &= \begin{cases} 0 & \text{if } R_{m,i,j} \geq c \\ N_{i,j} - R_{m,i,j} & \text{otherwise} \end{cases}
\end{aligned}$$

Essentially, it compensates for the difference between $N_{i,j}$ and $R_{m,i,j}$ when $R_{m,i,j} < c$. For $P_{m,c}(A, B)$ to be a well-defined kernel, it must have finitely many nonzero taps. This is true over c -saturating semirings, where only finitely many pairs (i, j) have $D_{m,i,j} \neq 0$, because for all sufficiently large (i, j) the multiplicity $R_{m,i,j}$ either saturates or already equals $N_{i,j}$.

More precisely, we claim that $D_{m,i,j} = 0$ whenever $i \geq c$ or $j \geq m+c$. Throughout, we use the conventions $\binom{n}{k} = 0$ for $k < 0$ and for $k > n \geq 0$.

Let us first check this claim on the axes (i.e., $i = 0$ or $j = 0$). They correspond to paths made of A 's only or of B 's only, and the multiplicity is 1 on both sides of the equation. When $j = 0$ and $i \geq 1$, $R_{m,i,0} = \binom{i-1}{0} = 1 = N_{i,0}$. When $i = 0$ and $j \geq m$, $R_{m,0,j} = \binom{j-m}{j-m} = 1 = N_{0,j}$. Thus $D_{m,i,j} = 0$ in both cases.

Off the axes, the multiplicity reaches the cap. First, when $i \geq c$ and $j \geq 1$, the

first summand of $R_{m,i,j}$ already suffices:

$$\begin{aligned} \mathbf{1}_{i>0} \binom{i+j-1}{j} + \mathbf{1}_{j \geq m} \binom{i+j-m}{j-m} &\geq \binom{i+j-1}{j} = \binom{i+j-1}{i-1} \\ &\geq \binom{i}{i-1} \\ &= i \geq c. \end{aligned}$$

Second, when $j \geq m+c$ and $i \geq 1$, the second summand already suffices:

$$\begin{aligned} \mathbf{1}_{i>0} \binom{i+j-1}{j} + \mathbf{1}_{j \geq m} \binom{i+j-m}{j-m} &\geq \binom{i+j-m}{j-m} = \binom{i+j-m}{i} \\ &\geq \binom{j-m+1}{1} \\ &= j-m+1 \geq c+1. \end{aligned}$$

In both cases $R_{m,i,j} \geq c$, so $D_{m,i,j} = 0$ by definition.

Together, we have shown that $D_{m,i,j} = 0$ whenever $i \geq c$ or $j \geq m+c$. Therefore,

$$P_{m,c}(A, B) = \bigoplus_{\substack{0 \leq i < c \\ 0 \leq j < m+c}} \langle D_{m,i,j} \rangle \otimes A^i \otimes B^j$$

is a finite kernel. Substituting this definition into the equation above, we can verify $F = \text{Rec}(A \oplus B, G)$ satisfies $F = (A \oplus B^m) \otimes F \oplus P_{m,c}(A, B) \otimes G$, and this fixed-point equation has a unique solution. Therefore

$$\boxed{\text{Rec}(A \oplus B, G) = \text{Rec}(A \oplus B^m, P_{m,c}(A, B) \otimes G).} \quad (19)$$

Two examples

When $c = 1$,

$$\begin{aligned} P_{m,1}(A, B) &= \bigoplus_{0 \leq j < m+1} \langle D_{m,0,j} \rangle \otimes B^j \\ &= \bigoplus_{0 \leq j < m+1} \left\langle \begin{cases} 0 & \text{if } \mathbf{1}_{j \geq m} \geq 1 \\ 1 - \mathbf{1}_{j \geq m} & \text{otherwise} \end{cases} \right\rangle \otimes B^j \\ &= \bigoplus_{0 \leq j < m} B^j \end{aligned}$$

and the dilation identity becomes

$$\text{Rec}(A \oplus B, G) = \text{Rec}\left(A \oplus B^m, \left(\bigoplus_{0 \leq j < m} B^j\right) \otimes G\right). \quad (20)$$

This is exactly the dilation identity for idempotent semirings in the paper.

For $c = 2$ and $m = 2$, the finite prefilter is

$$\begin{aligned}
P_{2,2}(A, B) &= \bigoplus_{\substack{0 \leq i < 2 \\ 0 \leq j < 4}} \langle D_{2,i,j} \rangle \otimes A^i \otimes B^j \\
&= \bigoplus_{0 \leq j < 2} B^j \oplus \bigoplus_{0 \leq j < 4} \left\langle \begin{cases} 0 & \text{if } R_{2,1,j} \geq 2 \\ N_{1,j} - R_{2,1,j} & \text{otherwise} \end{cases} \right\rangle \otimes A \otimes B^j \\
&= \bigoplus_{0 \leq j < 2} B^j \oplus \bigoplus_{0 \leq j < 4} \left\langle \begin{cases} 0 & \text{if } \mathbf{1}_{j \geq 2} \binom{j-1}{j-2} \geq 1 \\ N_{1,j} - 1 - \mathbf{1}_{j \geq 2} \binom{j-1}{j-2} & \text{otherwise} \end{cases} \right\rangle \otimes A \otimes B^j \\
&= \bigoplus_{0 \leq j < 2} B^j \oplus \bigoplus_{0 \leq j < 2} \langle N_{1,j} - 1 \rangle \otimes A \otimes B^j \\
&= I \oplus B \oplus A \otimes B
\end{aligned}$$

so

$$\text{Rec}(A \oplus B, G) = \text{Rec}(A \oplus B^2, (I \oplus B \oplus A \otimes B) \otimes G). \quad (21)$$

The extra $A \otimes B$ term is necessary because both paths $A \otimes B$ and $B \otimes A$ must be accounted for, and a 2-saturating semiring still distinguishes one copy from two. For longer mixed paths, once the new recurrence operator produces at least two copies, additional copies arising from different orderings no longer matter.

Example: arithmetic that counts only up to c

For $c \geq 1$, consider the values

$$\mathbb{N}_{\leq c} = \{0, 1, \dots, c\}$$

with

$$x \oplus_{\tau} y = \min(c, x + y), \quad x \otimes_{\tau} y = \min(c, xy).$$

This is a semiring, and it is c -saturating because adding one more copy after reaching c changes nothing. It is useful for problems that care how many paths exist but do not need the exact count once it reaches c .

The prefilter $P_{m,c}(A, B)$ grows with both c and m : the (rather conservative) bound above leaves at most $c \cdot (m + c)$ candidate pairs (i, j) , hence at most that many $A^i \otimes B^j$ terms. The transformation is therefore most attractive for small saturation caps, and is unlikely to be practical with, for example, saturating arithmetic over 32-bit integers.

Conclusion

To summarize, we have introduced three extensions of the dilation-like identities from the paper:

- When the recurrence is over a ring with an m -th root of unity, we have

$$\boxed{\text{Rec}(K, G) = \text{Rec}(L_m(K), P_m(K) \otimes G) .}$$

If the ring is moreover zero-divisor-free and the root is primitive, then $L_m(K)$ has stride ms , so this is a factor- m dilation.

- When the recurrence is over a ring of prime characteristic p , then, for any $q = p^h$ with $h \geq 1$, writing $K = \bigoplus_{i=1}^n K_i$ as a sum of singleton kernels, one for each tap of K , we have

$$\boxed{\text{Rec}(K, G) = \text{Rec}\left(\bigoplus_{i=1}^n K_i^q, \bigoplus_{i=0}^{q-1} K^i \otimes G\right)}$$

- When the recurrence is over a c -saturating semiring, we have

$$\boxed{\text{Rec}(A \oplus B, G) = \text{Rec}(A \oplus B^m, P_{m,c}(A, B) \otimes G) .}$$